

ANSWER KEY

Quiz 3

Course: CSC354 - Machine Learning

CLO-5 Develop a reasonable size project using appropriate machine learning technique.

Q1. Develop a Python program to implement a recommendation system using collaborative filtering on a movie rating dataset. The system should be able to predict the ratings for movies that users have not yet rated based on their past behavior and similar user behaviors. Your solution must include data preprocessing, model training with appropriate hyperparameters tuning, and evaluation of the model's performance using mean absolute error (MAE) metric.

Answer:

Algorithm: 1. Load movie rating dataset 2. Preprocess data by handling missing values and splitting into train-test sets 3. Implement collaborative filtering algorithm such as User-User or Item-Item based CF 4. Tune hyperparameters using cross-validation to optimize model performance 5. Train the model on training set 6. Predict ratings for test set 7. Evaluate model using MAE metric

Marks: 50

Q2. Develop a Python program to implement a recommendation system using collaborative filtering on a movie rating dataset. The system should be able to predict ratings for movies that users have not yet rated based on the preferences of similar users. Your implementation should include data preprocessing, model training with Singular Value Decomposition (SVD), and evaluation metrics such as Mean Absolute Error (MAE) or Root Mean Squared Error (RMSE).

Answer:

Algorithm: 1. Load movie rating dataset 2. Preprocess the data to handle missing values 3. Split the data into training and testing sets 4. Train an SVD model on the training set 5. Predict ratings for test set using the trained model 6. Calculate MAE or RMSE as evaluation metric

Marks: 50

Generated by IntelliMark - March 5, 2026

By: Hadia Anwar